

Video Game Programming: Exercises

E02

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Reactive characters

- Add to scene E01 some static obstacles and generate the navmesh
- Add a third person controller to move in the scene
- Download two characters (for the player and the enemies) and running animations
- Create an enemy AI so that it looks around to find the player
- Make the enemies run towards the player when looking in player's direction and player is less than 10m away
- Build the game for Windows or macOS

Audio UI

- Add to scene E02 2D and 3D audio sources for respectively music and sound effects
- Add a first-person controller to move in the scene
- Create a mixer to control volume level
- Design a responsive UI, inspired from recent games, to toggle music and/or sound effects on and off
- Add a setting button to open the UI
- Test the game on WebGL at different resolutions and aspect ratios

Lighting

- Add to scene E03 static and dynamic objects with different URP material settings
- Add a first-person controller to move in the scene
- Add several real-time, baked, and mixed lights of different types
- Bake the scene using different lighting modes and light probe configurations
- Test game quality with different settings and profile it on mobile